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| **Status** | Finished |
| **Started** | Saturday, 17 October 2026, 5:34 PM |
| **Completed** | Sunday, 18 October 2026, 8:41 PM |
| **Duration** | 1 day 3 hours |
| **Marks** | 3.00/3.00 |
| **Grade** | **3.75** out of 3.75 (**100**%) |

**Question 1**

Correct

Mark 1.00 out of 1.00

Which system is equivalent to this one, because it has been obtained applying Gaussian elimination?

$$\left\{\begin{array}{c} x+y+14\cdot z=6 \\ -8\cdot x-6\cdot y-113\cdot z=-46 \\ -6\cdot x-22\cdot y-69\cdot z=-44 \end{array}\right\} = ?$$

a. $\left\{\begin{array}{c} x-7\cdot y+14\cdot z=6 \\ 2\cdot y-z=2 \\ 7\cdot z=8 \end{array}\right\}$

b. $\left\{\begin{array}{c} x+y+14\cdot z=6 \\ 2\cdot y-z=2 \\ z=8 \end{array}\right\}$

c. $\left\{\begin{array}{c} x+y+14\cdot z=6 \\ 2\cdot y-z=2 \\ 7\cdot z=8 \end{array}\right\}$ ✓

d. $\left\{\begin{array}{c} x+y+14\cdot z=6 \\ 2\cdot y-3\cdot z=2 \\ 7\cdot z=8 \end{array}\right\}$

e. (non of the above)

To use the Gaussian elimination method, first we will find the augmented matrix of the system. In this case, the augmented matrix of the given system is

$$M0 = \left(\begin{array}{cccc} 1 & 1 & 14 & 6 \\ -8 & -6 & -113 & -46 \\ -6 & -22 & -69 & -44 \end{array}\right)$$

The Gaussian elimination method consists in converting, using elementary row operations, a general matrix to a row echelon matrix (that is, with zeros under the main diagonal). This transformation is not unique, and therefore, we could obtain different matrices depending on the operations we do.

In the following, we apply the Gaussian elimination method in its automatic form, since in this case we will not find any zero pivots that would force us to permute rows:

We create zeros under the diagonal of the first column:

row 2 = row 2 - M0(2,1) / M0(1,1) * row 1 = row 2 + (8) row 1
row 3 = row 3 - M0(3,1) / M0(1,1) * row 1 = row 3 + (6) row 1

$$M0 = \left(\begin{array}{cccc} 1 & 1 & 14 & 6 \\ -8 & -6 & -113 & -46 \\ -6 & -22 & -69 & -44 \end{array}\right) \rightarrow M1 = \left(\begin{array}{cccc} 1 & 1 & 14 & 6 \\ 0 & 2 & -1 & 2 \\ 0 & -16 & 15 & -8 \end{array}\right)$$

We create zeros under the diagonal of the second column:

row 3 = row 3 - M1(3,2) / M1(2,2) * row 2 = row 3 + (8) row 2

$$M1 = \left(\begin{array}{cccc} 1 & 1 & 14 & 6 \\ 0 & 2 & -1 & 2 \\ 0 & -16 & 15 & -8 \end{array}\right) \rightarrow M2 = \left(\begin{array}{cccc} 1 & 1 & 14 & 6 \\ 0 & 2 & -1 & 2 \\ 0 & 0 & 7 & 8 \end{array}\right)$$

If we convert the augmented matrix into a system of equations, we obtain

$$\left\{\begin{array}{c} x+y+14\cdot z=6 \\ 2\cdot y-z=2 \\ 7\cdot z=8 \end{array}\right\}$$

At this point, if the resulting system is one of the given options of the exercise, that would be the correct solution. Otherwise, what we can do is to find the solution of the system solving the echelon system and substitute this solution to the given systems, to find if any of the given system is an equivalent system (they have the same solution).

In this case, the solution of the system is: $\left\{x=1, y=\frac{11}{7}, z=\frac{8}{7}\right\}$

and therefore the correct answer to the exercise is: $\left\{\begin{array}{c} x+y+14\cdot z=6 \\ 2\cdot y-z=2 \\ 7\cdot z=8 \end{array}\right\}$

The correct answer is: $\left\{\begin{array}{c} x+y+14\cdot z=6 \\ 2\cdot y-z=2 \\ 7\cdot z=8 \end{array}\right\}$

**Question 2**

Correct

Mark 1.00 out of 1.00

We want to solve the following system of equations using Cramer's rule

$13 \cdot x - 5 \cdot z + 17 = 0,$
$2 \cdot x + 6 \cdot y - 2 \cdot z + 16 = 0,$
$2 \cdot x - 5 \cdot y + 11 \cdot z - 73 = 0,$

taking into account that the order of the variables considered in the solution vector is $(x,y,z)$.

Indicate the computations we must do to compute the value of x:

$$x = \frac{\begin{vmatrix} -17 & 0 & -5 \\ -16 & 6 & -2 \\ 73 & -5 & 11 \end{vmatrix}}{838}$$

Indicate the computations we must do to compute the value of y:

$$y = \frac{\begin{vmatrix} 13 & -17 & -5 \\ 2 & -16 & -2 \\ 2 & 73 & 11 \end{vmatrix}}{838}$$

Indicate the computations we must do to compute the value of z:

$$z = \frac{\begin{vmatrix} 13 & 0 & -17 \\ 2 & 6 & -16 \\ 2 & -5 & 73 \end{vmatrix}}{838}$$

Do the previous computations and give the solution of the system:

x= 1

y= -1

z= 6

FORMAT OF THE ANSWER: If any of the results is a fraction, we have to write it down as a fraction and not as a decimal. For example, 15/7, -1/21.

The first thing that we need to do to solve a system of equations using Cramer's rule, is to determine the coefficient matrix and constant vector. In this case, we have:

$A = \begin{pmatrix} 13 & 0 & -5 \\ 2 & 6 & -2 \\ 2 & -5 & 11 \end{pmatrix}$ and $b = \begin{pmatrix} -17 \\ -16 \\ 73 \end{pmatrix}$

Then, if we denote by $x_i$ the i-th component of the solution vector, we have:

$$x_i = \frac{det(C_1, C_2, ..., C_{i-1}, b, C_{i+1}, ..., C_n)}{det(A)}$$

where $C_i$ is the i-th column of matrix A. That is, to determine x, we will substitute the first column of matrix A by b, to determine y, we will substitute the second column of matrix A by b and to determine z, we will substitute the third column of A by b.

Therefore, since det(A)=838,

$$x=\frac{\left|\begin{pmatrix}-17 & 0 & -5\\ -16 & 6 & -2\\ 73 & -5 & 11\end{pmatrix}\right|}{838},\ y=\frac{\left|\begin{pmatrix}13 & -17 & -5\\ 2 & -16 & -2\\ 2 & 73 & 11\end{pmatrix}\right|}{838},\ z=\frac{\left|\begin{pmatrix}13 & 0 & -17\\ 2 & 6 & -16\\ 2 & -5 & 73\end{pmatrix}\right|}{838}.$$

If we perform the computations previously indicated, we obtain:

$$x=\frac{\left|\begin{pmatrix}-17 & 0 & -5\\ -16 & 6 & -2\\ 73 & -5 & 11\end{pmatrix}\right|}{838}=\frac{838}{838}=1$$

$$y=\frac{\left|\begin{pmatrix}13 & -17 & -5\\ 2 & -16 & -2\\ 2 & 73 & 11\end{pmatrix}\right|}{838}=\frac{-838}{838}=-1$$

$$z=\frac{\left|\begin{pmatrix}13 & 0 & -17\\ 2 & 6 & -16\\ 2 & -5 & 73\end{pmatrix}\right|}{838}=\frac{5028}{838}=6$$

**Question 3**

Correct

Mark 1.00 out of 1.00

Consider the following system of equations given in matrix form

$$\begin{pmatrix} 21 & -8 & -4 \\ 1 & 12 & 5 \\ 5 & 10 & 24 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 87 \\ 96 \\ 129 \end{pmatrix}.$$

List the operations that have been done below to transform the augmented matrix of the system to a row echelon matrix:

$$\begin{pmatrix} 21 & -8 & -4 & 87 \\ 1 & 12 & 5 & 96 \\ 5 & 10 & 24 & 129 \end{pmatrix} \rightarrow \text{row 2 = row 2 + } \boxed{-1/21} \cdot \text{row 1} \rightarrow \begin{pmatrix} 21 & -8 & -4 & 87 \\ 0 & \frac{260}{21} & \frac{109}{21} & \frac{643}{7} \\ 5 & 10 & 24 & 129 \end{pmatrix}$$

$$\begin{pmatrix} 21 & -8 & -4 & 87 \\ 0 & \frac{260}{21} & \frac{109}{21} & \frac{643}{7} \\ 5 & 10 & 24 & 129 \end{pmatrix} \rightarrow \text{row 1 = } \boxed{1/21} \cdot \text{row 1} \rightarrow \begin{pmatrix} 1 & -\frac{8}{21} & -\frac{4}{21} & \frac{29}{7} \\ 0 & \frac{260}{21} & \frac{109}{21} & \frac{643}{7} \\ 5 & 10 & 24 & 129 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -\frac{8}{21} & -\frac{4}{21} & \frac{29}{7} \\ 0 & \frac{260}{21} & \frac{109}{21} & \frac{643}{7} \\ 5 & 10 & 24 & 129 \end{pmatrix} \rightarrow \text{row 3 = row 3 + } \boxed{-5} \cdot \text{row 1} \rightarrow \begin{pmatrix} 1 & -\frac{8}{21} & -\frac{4}{21} & \frac{29}{7} \\ 0 & \frac{260}{21} & \frac{109}{21} & \frac{643}{7} \\ 0 & \frac{250}{21} & \frac{524}{21} & \frac{758}{7} \end{pmatrix}$$

$$\begin{pmatrix} 1 & -\frac{8}{21} & -\frac{4}{21} & \frac{29}{7} \\ 0 & \frac{260}{21} & \frac{109}{21} & \frac{643}{7} \\ 0 & \frac{250}{21} & \frac{524}{21} & \frac{758}{7} \end{pmatrix} \rightarrow \text{row 3 = row 3 + } \boxed{-25/26} \cdot \text{row 2} \rightarrow \begin{pmatrix} 1 & -\frac{8}{21} & -\frac{4}{21} & \frac{29}{7} \\ 0 & \frac{260}{21} & \frac{109}{21} & \frac{643}{7} \\ 0 & 0 & \frac{519}{26} & \frac{519}{26} \end{pmatrix}$$

Give the solution of the system of equations:

x= 7

y= 7

z= 1

FORMAT OF THE ANSWER: If any of the results is a fraction, we have to write it down as a fraction and not as a decimal. For example, 15/7, -1/21.

We explain below which are the operations that have been done:

- note that in the first step, we have obtained a zero in position (2,1) of the matrix (second row, first column). To do so, we have done the following operation
  row 2 = row 2 + c · row 1
  If we take the coefficients that we have in the first column, and we substitute them in the previous equation, we obtain:
  0 = 1 + c · 21 → c = - 1 / 21

- in the second step, we have obtained a 1 in position (1,1) of the matrix (first row, first column). To do so, we have done the following operation
  row 1 = c · row 1
  If we take the coefficients that we have in the first column, and we substitute them in the previous equation, we obtain:
  1 = c · 21 → c = 1 / 21

- in the third step, we have obtained a zero in position (3,1) of the matrix (third row, first column). To do so, we have done the following operation
  row 3 = row 3 + c · row 1
  If we take the coefficients that we have in the first column, and we substitute them in the previous equation, we obtain:

$0 = 5 + c \cdot 1 \rightarrow c = -5$

- finally, in the last step, we have obtained a zero in position (3,2) of the matrix (third row, second column). To do so, we have done the following operation
  row 3 = row 3 + c · row 2
  If we take the coefficients that we have in the second column, and we substitute them in the previous equation, we obtain:
  $0 = \frac{250}{21} + c \cdot \frac{260}{21} \rightarrow c = -(\frac{250}{21})/(\frac{260}{21}) = -\frac{25}{26}$

The solution of the system is then found by observing that the last matrix is row echelon, and therefore, the solution can be found using backward substitution (finding first z, then y and finally x).